

Cloud Computing TASK – 5

- 1. Pick a use case from an app you use daily (for example: Instagram photo sharing, Zomato food ordering, or Flipkart online shopping) and briefly describe the main technical requirements for hosting this app online.**

Ans : Instagram is a social media application where users upload photos and videos, follow other users, like and comment on posts, and send messages.

Main Technical Requirements for Hosting the App Online:

1. Cloud Infrastructure

- Virtual servers or cloud platforms to run the application backend.
- Auto-scaling resources to handle millions of users during high traffic periods.

2. Database Storage

- A database to store user profiles, posts, comments, likes, and messages.
- A scalable database system to handle large amounts of user activity.

3. File Storage

- Cloud storage for storing uploaded photos and videos.
- Content Delivery Network (CDN) to deliver images and videos quickly worldwide.

4. Networking

- High-speed internet connectivity, load balancers, and secure communication using HTTPS.

- Domain name and DNS services to make the app accessible online.

5. Security

- User authentication and authorization systems.
- Encryption of user data and secure access controls using IAM policies.
- Protection against attacks such as DDoS and unauthorized access.

**2. Create a table comparing public, private, and hybrid cloud deployment models for your chosen use case. Include at least 4 factors: cost, security, scalability, and ease of maintenance.

Hint: Think about how each model would handle user data, sudden traffic spikes, and updates for your app scenario.**

Ans :

Factor	Public Cloud	Private Cloud	Hybrid Cloud
Cost	Lower cost because resources are shared with other customers. Suitable for handling large user bases without	Higher cost because the company must manage dedicated infrastructure, servers, and maintenance.	Medium cost because critical systems use private resources while general workloads use public cloud services.

Factor	Public Cloud	Private Cloud	Hybrid Cloud
	owning hardware.		
Security	Good security provided by the cloud provider, but user data requires proper IAM controls and encryption to prevent unauthorized access.	Highest control over security because user data and systems run on dedicated infrastructure. Useful for storing sensitive information.	Provides strong security by keeping sensitive user data in private systems while using public cloud for less sensitive tasks.
Scalability	Highly scalable; can quickly add resources during sudden traffic spikes such as viral posts or live events.	Limited scalability because expansion requires purchasing and configuring additional hardware.	Flexible scalability; public cloud can handle sudden increases in traffic while private infrastructure manages core services.

3. Write a short paragraph explaining which cloud deployment model you would recommend for your chosen use case and

why. Support your answer with at least two points from your comparison table.

Ans : For the Instagram photo-sharing app, I would recommend a hybrid cloud deployment model. It provides a balance between security, scalability, and cost efficiency. Sensitive user data such as account details and private messages can be stored in a private cloud environment for better control and security. At the same time, public cloud resources can be used to store and deliver photos/videos and handle sudden traffic spikes when many users upload or view content. According to the comparison table, hybrid cloud offers better scalability than a private cloud and provides stronger data protection than relying only on a public cloud, making it a suitable choice for a large-scale social media application.

4. Imagine your chosen app becomes extremely popular overnight (like during IPL finals or a big sale). Describe one advantage and one disadvantage of using a hybrid cloud model in this situation.

Ans : For the Instagram photo-sharing app, if it becomes extremely popular overnight, a hybrid cloud model would provide the following advantage and disadvantage:

Advantage:

A hybrid cloud can handle sudden traffic spikes effectively by using public cloud resources to automatically scale up servers, storage, and bandwidth. For example, during a major event, millions of users uploading and viewing photos can be supported without affecting the private systems storing sensitive user information.

Disadvantage:

Managing a hybrid cloud environment is more complex because the company must maintain both private and public cloud systems.

Proper configuration, data synchronization, and security policies are required to ensure that user data remains protected while workloads move between environments.

5. Use ChatGPT or any AI tool to generate a summary of when to use public, private, and hybrid clouds for popular Indian apps (such as Paytm, IRCTC, BookMyShow). Paste the summary and highlight one insight you found surprising.

Ans : Summary: When to Use Public, Private, and Hybrid Clouds for Popular Indian Apps

- **Public Cloud:**

Public cloud is suitable for apps that need high scalability, quick deployment, and cost efficiency. Apps with millions of users and unpredictable traffic, such as BookMyShow during movie releases or sports events, can use public cloud resources to handle sudden increases in users, ticket searches, and bookings.

- **Private Cloud:**

Private cloud is useful for applications that handle highly sensitive data and require strict control over security and compliance. Financial applications such as Paytm may use private cloud environments for critical payment-related systems where data protection and regulatory requirements are important.

- **Hybrid Cloud:**

Hybrid cloud combines the advantages of public and private clouds. Apps such as IRCTC can benefit from hybrid cloud because they need strong security for passenger information while also requiring high scalability during festival seasons, holidays, and peak booking periods.